Strava Fitness Report

Introduction

Fitness tracking platforms collect large amounts of data related to users' physical activity, sleep patterns, and health metrics. Analyzing this data helps understand how users engage with fitness tracking systems and monitor their daily health activities.

This project analyzes fitness tracker data to explore patterns in user behavior, including daily steps, calories burned, activity intensity, sleep duration, and weight records. Python is used for data cleaning and exploratory data analysis, while MySQL is used for structured queries. An interactive Power BI dashboard is created to visualize the insights and highlight trends in user activity and engagement.

Problem Statement

- Fitness tracking platforms collect large amounts of data related to users' physical activity, sleep patterns, and health metrics.
- However, understanding how users engage with these fitness tracking devices requires proper data analysis.
- The main problem is to analyze user engagement and activity patterns using fitness tracker data.
- The dataset includes information about users' daily activity, sleep records, and weight logs.
- This project aims to examine how actively users track their health and fitness activities.
- It also analyzes whether users spend more time in active or sedentary behavior.
- The project investigates patterns such as average daily steps, calories burned, and sleep duration.
- Another objective is to explore relationships between physical activity and sleep habits.
- The insights generated from this analysis can help understand user behavior and engagement with fitness tracking platforms.

Objectives

The main objectives of this project are:

- Understand how users interact with the fitness tracking platform and measure overall user engagement.
- Analyze users' daily activity levels using metrics such as total steps, calories burned, and active minutes.
- Identify patterns in user behavior, including how activity levels vary across different days of the week.
- Evaluate the balance between active time and sedentary time to understand user lifestyle patterns.
- Examine sleep tracking data to understand users' sleep duration and sleep habits.
- Determine how many users actively track additional health metrics such as weight and BMI.
- Explore the relationship between physical activity and other health indicators like sleep and calories burned.

- Generate insights that can help fitness platforms improve user engagement and encourage healthier lifestyle habits.
- Provide data-driven insights that could help companies design better fitness features and personalized health recommendations.

Dataset Description

The dataset contains information related to users' fitness tracking activities, including daily physical activity, sleep patterns, and weight records. The data helps analyze user engagement, activity levels, calories burned, sleep duration, and health metrics recorded by fitness tracking devices.

Activity Dataset

Column Name	Description
Id	**Unique identifier for each user**
ActivityDate	**Date when the activity was recorded**
TotalSteps	**Total number of steps taken in a day**
TotalDistance	**Total distance covered by the user**
VeryActiveMinutes	**Minutes spent in intense physical activity**
FairlyActiveMinutes	**Minutes spent in moderate activity**
LightlyActiveMinutes	**Minutes spent in light activity**
SedentaryMinutes	**Minutes spent inactive**
Calories	**Total calories burned in a day**
DayOfWeek	**Day of the week of the recorded activity**
Month	**Month of the recorded activity**

Sleep Dataset

Column Name	Description
Id	Unique identifier for each user
SleepDay	Date when sleep data was recorded
TotalSleepRecords	Number of sleep records logged
TotalMinutesAsleep	Total minutes the user was asleep
TotalTimeInBed	Total time the user spent in bed

Weight Dataset

Column Name	Description
Id	Unique identifier for each user
Date	Date when weight was recorded
WeightKg	User weight in kilograms
WeightPounds	User weight in pounds
BMI	Body Mass Index of the user
IsManualReport	Indicates whether weight was manually reported
LogId	Unique identifier for each weight record

Tools Used

The following tools were used in this project:

- **Python**: Data cleaning and exploratory data analysis
- **SQL (MySQL)**: Querying data and calculating KPIs
- **Power Bi** : Dashboard creation and data visualization

Python EDA Summary

Exploratory Data Analysis (EDA) was performed using Python to understand, clean, and transform the fitness tracking dataset before conducting further analysis and visualization.

Data Preprocessing & Feature Engineering

During the EDA process, additional columns were created from the activity date field to enable time-based analysis of user behavior:

• **DayOfWeek**
Extracted from the activity date to analyze user activity patterns across different days of the week.

• **Month**
Derived from the activity date to study monthly activity trends and identify any variations in user engagement over time.

These newly created columns helped in identifying weekly activity patterns and understanding how user activity varies across different days.

Handling Missing Values

• The **daily activity dataset** did not contain missing values and was ready for analysis.
• In the **weight dataset**, the **Fat column** contained a large number of missing values and was removed because it was not suitable for meaningful analysis.

• Other columns such as **BMI, WeightKg, and WeightPounds** were retained as they contained valid health-related information.

EDA Findings Summary

• Users record their daily activity regularly, providing detailed information about steps, calories burned, and activity intensity levels.
• Most users spend significantly more time in **sedentary activities** compared to active minutes, indicating relatively inactive daily routines.
• There is a clear positive relationship between **total steps and calories burned**, showing that higher physical activity results in greater energy expenditure.
• Sleep tracking data reveals variations in **sleep duration across users**, highlighting different sleep habits.
• Weight tracking data is recorded by fewer users compared to activity tracking, indicating lower engagement with health metrics such as weight and BMI.

SQL Queries & Analysis

SQL was used to compute key user engagement metrics and validate insights obtained from the Python exploratory data analysis. The following queries focus on analyzing user activity levels, sleep behavior, and health tracking patterns in the Strava fitness dataset.

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1. Total Number of Users

Objective:
To determine the total number of unique users in the dataset.

```
SELECT COUNT(DISTINCT Id) AS total_users

FROM dailyactivity;
```

Analysis:
This query calculates the total number of unique users who recorded activity data. It helps establish the size of the user base represented in the dataset.

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2. Average Daily Steps

Objective:
To measure the average number of steps taken by users per day.

```
SELECT AVG(TotalSteps) AS avg_daily_steps

FROM dailyactivity;
```

Analysis:
This metric provides an overall view of user physical activity levels and helps evaluate whether users maintain an active lifestyle.

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3. Most Active Users

Objective:
To identify the most active users based on their average step count.

```
SELECT Id, AVG(TotalSteps) AS avg_steps

FROM dailyactivity
```

```
GROUP BY Id

ORDER BY avg_steps DESC

LIMIT 10;
```

Analysis:
This query highlights the top users with the highest average number of steps, helping identify highly engaged or physically active users.

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4. Average Activity Minutes

Objective:
To compare different levels of activity intensity among users.

```
SELECT

AVG(VeryActiveMinutes) AS avg_very_active,

AVG(FairlyActiveMinutes) AS avg_fairly_active,

AVG(LightlyActiveMinutes) AS avg_light_active,

AVG(SedentaryMinutes) AS avg_sedentary

FROM dailyactivity;
```

Analysis:
This analysis reveals how users distribute their time across different activity levels and helps identify whether users spend more time being active or sedentary.

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5. Average Steps by Day of Week

Objective:
To analyze user activity trends across different days of the week.

```
SELECT DayOfWeek, AVG(TotalSteps) AS avg_steps

FROM dailyactivity

GROUP BY DayOfWeek

ORDER BY avg_steps DESC;
```

Analysis:
This query identifies which days users are most active and helps understand weekly activity patterns.

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6. Calories Burned vs Steps

Objective:
To examine the relationship between physical activity and calories burned.

```
SELECT

AVG(TotalSteps) AS avg_steps,

AVG(Calories) AS avg_calories

FROM dailyactivity;
```

Analysis:
This helps validate the relationship between step count and calorie expenditure, showing that higher physical activity generally leads to higher calories burned.

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7. Average Sleep Duration

Objective:
To calculate the average sleep duration recorded by users.

```
SELECT AVG(TotalMinutesAsleep) AS avg_sleep_minutes

FROM sleepday;
```

Analysis:
This metric provides insight into user sleep habits and overall sleep duration.

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8. Activity vs Sleep

Objective:
To analyze the relationship between physical activity and sleep duration.

```
SELECT

a.Id,

AVG(a.TotalSteps) AS avg_steps,

AVG(s.TotalMinutesAsleep) AS avg_sleep

FROM dailyactivity a

JOIN sleepday s

ON a.Id = s.Id

GROUP BY a.Id;
```

Analysis:
This query combines activity and sleep data to explore whether users who are more active also tend to have longer sleep durations.

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9. Users Who Log Weight

Objective:
To determine how many users actively track their weight.

```
SELECT COUNT(DISTINCT Id) AS users_logging_weight

FROM weightloginfo;
```

Analysis:
This helps measure engagement with health tracking features beyond physical activity.

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10. Average BMI of Users

Objective:
To calculate the average BMI of users in the dataset.

```
SELECT AVG(BMI) AS avg_bmi

FROM weightloginfo;
```

Analysis:
This provides an overview of the general health condition of users based on BMI values.

Dashboard Overview

1. Dashboard KPIs
The top section of the dashboard displays key performance indicators (KPIs) that summarize overall user activity and engagement levels in the fitness dataset:

- **Total Users:** Total number of unique users recorded in the dataset
- **Average Daily Steps:** Average number of steps taken by users per day
- **Average Calories Burned:** Average calories burned by users during daily activities
- **Average Sleep Duration:** Average number of minutes users spend sleeping

These KPIs provide a quick overview of overall fitness activity and user engagement.

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2. Average Steps by Day of Week
The **Average Steps by Day of Week** chart shows how user activity varies across different days of the week.

- Certain days show higher average step counts compared to others
- Activity levels fluctuate throughout the week depending on user routines
- Helps identify the days when users tend to be most physically active

This visualization helps understand weekly activity patterns among users.

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3. Steps vs Calories Analysis
The **Steps vs Calories** scatter plot shows the relationship between the number of steps taken and calories burned.

- Higher step counts generally correspond to higher calories burned
- Indicates a strong relationship between physical activity and energy expenditure
- Helps confirm that increased movement leads to higher calorie consumption

This chart highlights the direct connection between user activity levels and calorie burn.

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4. Average Calories by Day of Week
The **Average Calories by Day of Week** chart illustrates how calorie expenditure changes throughout the week.

- Calorie burn trends closely follow activity patterns
- Days with higher average steps tend to show higher calorie expenditure
- Helps identify days when users are more physically active

This visualization reinforces the relationship between physical activity and calorie burn.

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5. Sleep vs Time in Bed Analysis
The **Sleep vs Time in Bed** scatter plot shows the relationship between the total time users spend in bed and the actual minutes they spend asleep.

• Users who spend more time in bed generally tend to sleep longer
• The relationship helps evaluate sleep efficiency
• Highlights variations in sleep patterns across different users

This visualization helps analyze user sleep behavior and sleeping habits.

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6. BMI by Users
The **BMI by Users** chart displays the Body Mass Index values recorded for different users.

• Shows variation in BMI values across users
• Provides an overview of user health conditions
• Helps identify differences in body composition among individuals

This chart provides insight into users' health metrics.

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7. Average Sleep Duration by User
The **Average Sleep Duration by User** chart compares average sleep duration across different users.

• Some users consistently record higher sleep durations than others
• Sleep patterns vary significantly between individuals
• Helps identify users with shorter or longer sleep durations

This visualization highlights differences in user sleep habits.

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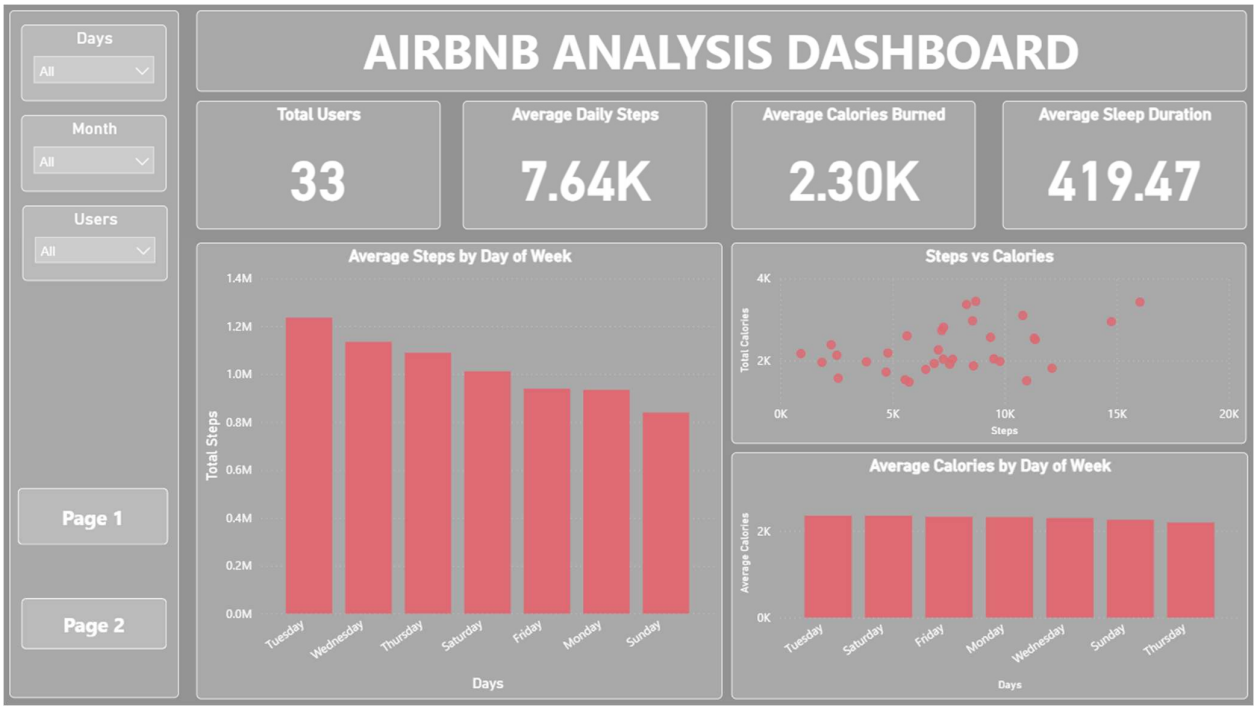
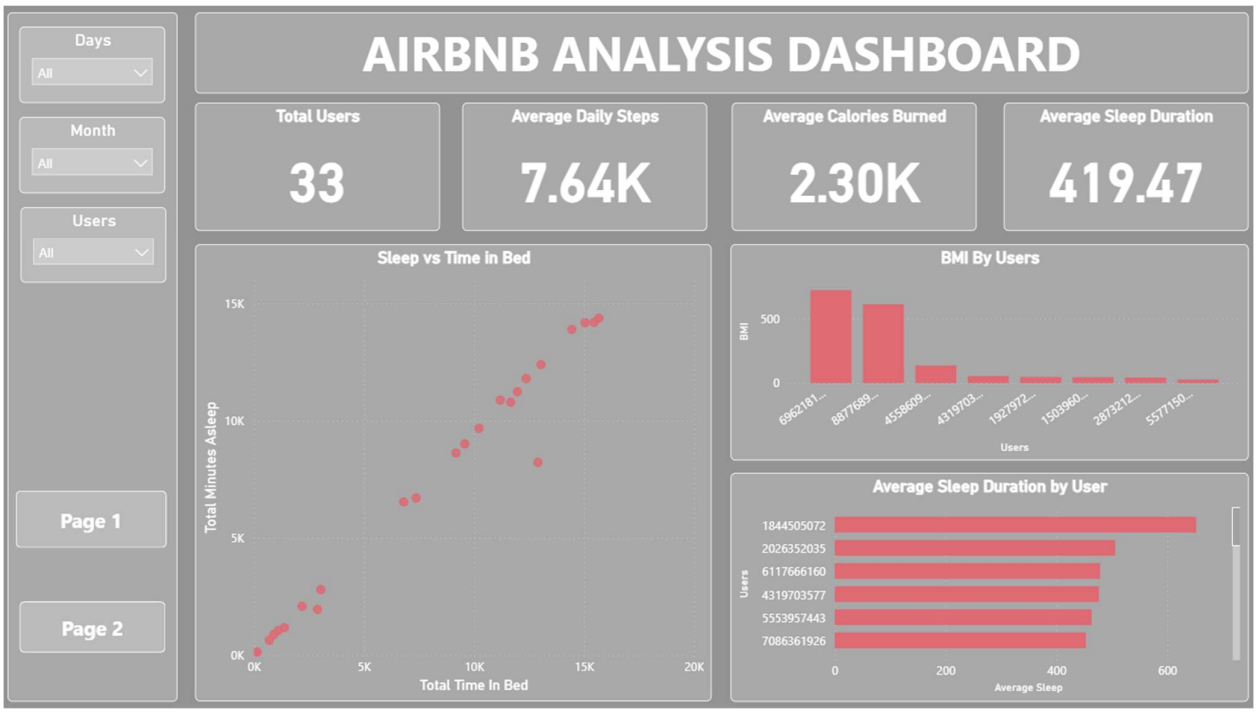
8. Interactive Filters
The dashboard includes interactive slicers that allow users to dynamically explore the data.

• **Days:** Filters activity based on the day of the week
• **Month:** Allows analysis of activity patterns across different months
• **Users:** Enables exploration of activity data for specific users

These filters allow users to explore trends and analyze user behavior interactively.





Key Insights

• User activity levels vary across different days of the week, indicating weekly behavioral patterns.

• There is a strong positive relationship between steps taken and calories burned, confirming that increased activity leads to higher energy expenditure.

• Users spend significantly more time in sedentary behavior compared to active minutes, suggesting that many users maintain relatively inactive lifestyles.

• Sleep patterns vary across users, with some individuals consistently achieving longer sleep durations.

• The relationship between time spent in bed and actual sleep duration indicates varying levels of sleep efficiency among users.

• Weight and BMI data provide additional insights into user health metrics, although fewer users actively track their weight compared to activity data.

Recommendations

Based on the analysis and insights derived from Python EDA, SQL queries, and the Power BI dashboard, the following recommendations are proposed to improve user engagement and promote healthier lifestyle habits on fitness tracking platforms:

• **Encourage consistent daily activity**
Users should be encouraged to maintain regular physical activity through reminders, goal-setting features, and step challenges to increase daily movement.

• **Promote active lifestyle habits**
Since users spend a significant amount of time in sedentary activities, fitness platforms can introduce activity alerts or reminders to encourage users to move more frequently.

- **Provide personalized fitness insights**
Using activity and sleep data, personalized recommendations can be generated to help users improve their exercise routines and health habits.

- **Improve sleep tracking awareness**
Users can benefit from insights on sleep efficiency and sleep duration, encouraging better sleep routines and healthier lifestyle patterns.

- **Encourage health metric tracking**
Since fewer users log weight data compared to activity data, platforms can promote the use of weight and BMI tracking features to provide a more complete picture of personal health.

Conclusions

This project successfully analyzed fitness tracking data using Python, SQL, and Power BI to understand user activity patterns, sleep behavior, and health engagement. Through exploratory data analysis and structured querying, key trends related to physical activity levels, calorie expenditure, and sleep patterns were identified.

The analysis revealed that user activity levels vary across different days of the week and that higher step counts are associated with increased calorie expenditure. It also highlighted that users spend a large portion of their time in sedentary activities, suggesting opportunities to encourage more active lifestyles. Additionally, variations in sleep duration and BMI values provide insights into user health and wellness patterns.

The Power BI dashboard effectively visualizes these insights, enabling easy exploration of user activity, sleep habits, and health metrics. Overall, this project demonstrates how data-driven analysis can help understand user engagement with fitness tracking platforms and provide actionable insights to promote healthier lifestyles and improved health monitoring.